First Order Logic

This Lecture

Last time we talked about propositional logic, a logic on simple statements.

This time we will talk about first order logic, a logic on quantified statements.

First order logic is much more expressive than propositional logic.

The topics on first order logic are:

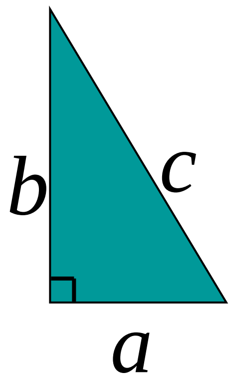
- Quantifiers
- Negation
- Multiple quantifiers
- Arguments of quantified statements
- (Optional) Important theorems, applications

Limitation of Propositional Logic

Propositional logic – logic of simple statements

$$\neg, \wedge, \vee, \longrightarrow, \longleftrightarrow$$

How to formulate Pythagoreans' theorem using propositional logic?



How to formulate the statement that there are infinitely many primes?

Predicates

Predicates are propositions (i.e. statements) with variables

Example: $P(x,y) ::= x + 2 = y$

$x = 1$ and $y = 3$: $P(1,3)$ is true

$x = 1$ and $y = 4$: $P(1,4)$ is false

$\neg P(1,4)$ is true

When there is a variable, we need to specify what to put in the variables.

The **domain** of a variable is the set of all values that may be substituted in place of the variable.

Set

To specify the domain, we often need the concept of a **set**.

Roughly speaking, a set is just a collection of objects.

Some examples

R	Set of all real numbers
Z	Set of all integers
Q	Set of all rational numbers

Given a set, the (only) important question is whether an element belongs to it.

$x \in A$ means that x is an element of *A* (pronounce: x in *A*)

$x \notin A$ means that x is not an element of *A* (pronounce: x not in *A*)

Sets can be defined explicitly: e.g. {1,2,4,8,16,32,…}, {CSC1130,CSC2110,…}

Truth Set

Sometimes it is inconvenient or impossible to define a set explicitly.

Sets can be defined by a predicate

Given a predicate P(x) where x has domain D, the truth set of P(x) is the set of all elements of D that make P(x) true.

$$\{x \in D \mid P(x)\}$$

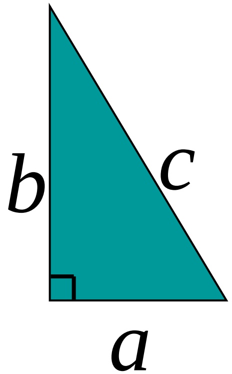
e.g. Let P(x) be "x is the square of a number",
and the domain D of x is the set of positive integers.
Then the truth set is the set of all positive integers which are the square of a number.

e.g. Let P(x) be "x is a prime number",
and the domain D of x is the set of positive integers.
Then the truth set is the set of all positive integers which are prime numbers.

The Universal Quantifier

The universal quantifier $\forall x$ for ALL x

Example: $\forall x \in Z \ \forall y \in Z, \ x + y = y + x.$



Pythagorean's theorem

$\forall$ right – angled triangle $a^2 + b^2 = c^2$

Example: $\forall x \quad x^2 \geq x$

This statement is true if the domain is Z, but not true if the domain is R.

The truth of a statement depends on the domain.

The Existential Quantifier

$\exists y$ There EXISTS some y

e.g. $\exists y, y^2 = y$

The truth of a statement depends on the domain.

$$\forall x \exists y.\ x < y$$

Domain	Truth value
integers $\mathbb{Z}$	T
positive integers $\mathbb{Z}^+$	T
negative integers $\mathbb{Z}^-$	F
negative reals $\mathbb{R}^-$	T

Translating Mathematical Theorem

Fermat (1637): If an integer n is greater than 2,

then the equation $a^n + b^n = c^n$ has no solutions in non-zero integers a, b, and c.

$$\forall n > 2 \ \forall a \in Z^+ \ \forall b \in Z^+ \ \forall c \in Z^+ \quad a^n + b^n \neq c^n$$

Andrew Wiles (1994) http://en.wikipedia.org/wiki/Fermat's_last_theorem

Translating Mathematical Theorem

Goldbach's conjecture: Every even number is the sum of two prime numbers.

Suppose we have a predicate prime(x) to determine if x is a prime number.

$$\forall n \in Z \ \text{even}(n) \rightarrow$$

$$\exists p \in Z \ \exists q \in Z \ \text{prime}(p) \wedge \text{prime}(q) \wedge (p+q = n)$$

How to write prime(p)?

$$\text{prime}(p) \ :=$$

$$(p > 1) \wedge (\forall a \in Z \ \forall b \in Z \ ((a > 1) \wedge (b > 1) \rightarrow a \cdot b \neq p))$$

- Quantifiers
- Negation
- Multiple quantifiers
- Arguments of quantified statements
- (Optional) Important theorems, applications

Negations of Quantified Statements

Everyone likes football.

What is the negation of this statement?

Not everyone likes football = There exists someone who doesn't like football.

$$\neg \forall x P(x) \equiv \exists x \neg P(x)$$

(generalized) DeMorgan's Law

Say the domain has only three values.

$$\begin{aligned}\neg \forall x P(x) &\equiv \neg(P(1) \wedge P(2) \wedge P(3)) \\ &\equiv \neg(P(1) \wedge P(2)) \vee \neg P(3)) \\ &\equiv \neg P(1) \vee \neg P(2) \vee \neg P(3) \quad \equiv \exists x \neg P(x)\end{aligned}$$

The same idea can be used to prove it for any number of variables, by mathematical induction.

Negations of Quantified Statements

There is a plant that can fly.

What is the negation of this statement?

Not exists a plant that can fly = every plant cannot fly.

$$\neg \exists x P(x) \equiv \forall x \neg P(x)$$

(generalized) DeMorgan's Law

Say the domain has only three values.

$$\begin{aligned}
\neg \exists x P(x) &\equiv \neg (P(1) \vee P(2) \vee P(3)) \\
&\equiv \neg (P(1) \vee P(2)) \wedge \neg P(3)) \\
&\equiv \neg P(1) \wedge \neg P(2) \wedge \neg P(3) \\
&\equiv \forall x \neg P(x)
\end{aligned}$$

The same idea can be used to prove it for any number of variables, by mathematical induction.

- Quantifiers
- Negation
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Order of Quantifiers

There is an anti-virus program killing every computer virus.

How to interpret this sentence?

For every computer virus, there is an anti-virus program that kills it.

$$\forall V \ \ \exists P, \mathsf{kill}(P, V)$$

- For every attack, I have a defense:
- against **MYDOOM**, use Defender
- against **ILOVEYOU**, use Norton
- against **BABLAS**, use Zonealarm ...

?? is expensive!

Order of Quantifiers

There is an anti-virus program killing every computer virus.

How to interpret this sentence?

There is one single anti-virus program that kills all computer viruses.

$$\exists P \ \forall V, \mathsf{kill}(P, V)$$

I have *one* defense good against every attack.

Example: P is CSE-antivirus,
protects against *ALL* viruses

That's much better!

Order of quantifiers is very important!

Order of Quantifiers

Let's say we have an array A of size 6x6.

$$\forall \text{ row } x \; \exists \text{ column } y \quad A[x,y] = 1$$

1					
	1	1		1	
		1			
		1		1	
			1		
		1			

Then this table satisfies the statement.

Order of Quantifiers

Let's say we have an array A of size 6x6.

$$\exists \text{ row } x \; \forall \text{ column } y \quad A[x, y] = 1$$

1					
	1	1		1	
		1			
		1		1	
			1		
		1			

But if the order of the quantifiers are changes,

then this table no longer satisfies the new statement.

Order of Quantifiers

Let's say we have an array A of size 6x6.

$$\exists \text{ row } x \ \forall \text{ column } y \quad A[x,y] = 1$$

1	1	1	1	1	1

To satisfy the new statement, there must be a row with all ones.

Questions

Are these statements equivalent?

$$\forall \text{ row } x \; \forall \text{ column } y \quad A[x,y] = 1$$

$$\forall \text{ column } y \; \forall \text{ row } x \quad A[x,y] = 1$$

Are these statements equivalent?

$$\exists \text{ row } x \; \exists \text{ column } y \quad A[x,y] = 1$$

$$\exists \text{ column } y \; \exists \text{ row } x \quad A[x,y] = 1$$

Yes, in general, you can change the order of two "foralls", and you can change the order of two "exists".

More Negations

There is an anti-virus program killing every computer virus.

$$\exists P \ \forall V, \mathsf{kill}(P, V)$$

What is the negation of this sentence?

$$\neg(\exists P \ \forall V, \mathsf{kill}(P, V))$$

$$\equiv \forall P \ \neg(\forall V, \mathsf{kill}(P, V))$$

$$\equiv \forall P \exists V \neg \mathsf{kill}(P, V)$$

For every program, there is some virus that it can not kill.

Exercises

1. There is a smallest positive integer.

2. There is no smallest positive real number.

3. There are infinitely many prime numbers.

Exercises

1. There is a smallest positive integer.

$$\exists s \in Z^+ \ \forall x \in Z^+ \ s \leq x$$

2. There is no smallest positive real number.

$$\forall r \in R^+ \ \exists x \in R^+ \ x < r$$

3. There are infinitely many prime numbers.

$$\exists p \in Z \ \ prime(p) \wedge$$
$$\forall p \in Z \ \ prime(p) \ \rightarrow \ \exists q \in Z \ \ prime(q) \wedge (q > p)$$

- Quantifiers
- Negation
- Multiple quantifiers
- **Arguments of quantified statements**
- (Optional) Important theorems, applications

Predicate Calculus Validity

Propositional validity

$$(A \rightarrow B) \vee (B \rightarrow A)$$

True ***no matter what*** the truth values of *A* and *B* are

Predicate calculus validity

$$\forall z\,[Q(z) \wedge P(z)] \rightarrow [\forall x.Q(x) \wedge \forall y.P(y)]$$

True ***no matter what***

- the Domain is,
- or the predicates are.

That is, logically correct, independent of the specific content.

Arguments with Quantified Statements

Universal instantiation:

$$\begin{array}{l} \forall x, P(x) \\ \therefore\ P(a) \end{array}$$

Universal modus ponens:

$$\begin{array}{l} \forall x, P(x) \rightarrow Q(x) \\ P(a) \\ \therefore\ Q(a) \end{array}$$

Universal modus tollens:

$$\begin{array}{l} \forall x, P(x) \rightarrow Q(x) \\ \neg Q(a) \\ \therefore\ \neg P(a) \end{array}$$

Universal Generalization

valid rule

$$\frac{A \rightarrow R(c)}{A \rightarrow \forall x.R(x)}$$

providing c is independent of *A*

Informally, if we could prove that R(c) is true for an arbitrary c (in a sense, c is a "variable"), then we could prove the for all statement.

e.g. given any number c, 2c is an even number

=> for all x, 2x is an even number.

Remark: Universal generalization is often difficult to prove, we will introduce mathematical induction to prove the validity of for all statements.

Valid Rule?

$$\forall z\, [Q(z) \lor P(z)] \rightarrow [\forall x.Q(x) \lor \forall y.P(y)]$$

Proof: Give ***countermodel***, where

$\forall z\, [Q(z) \lor P(z)]$ is true,

but $\forall x.Q(x) \lor \forall y.P(y)$ is false.

Find a domain, and a predicate.

In this example, let domain be integers,

$Q(z)$ be true if z is an even number, i.e. Q(z)=even(z)

$P(z)$ be true if z is an odd number, i.e. P(z)=odd(z)

Then $\forall z\, [Q(z) \lor P(z)]$ is true, because every number is either even or odd.

But $\forall x.Q(x)$ is not true, since not every number is an even number.

Similarly $\forall y.P(y)$ is not true, and so $\forall x.Q(x) \lor \forall y.P(y)$ is not true.

Valid Rule?

$$\forall z \in D \quad [Q(z) \wedge P(z)] \rightarrow [\forall x \in D\ Q(x) \wedge \forall y \in D\ P(y)]$$

Proof: Assume $\forall z\ [Q(z) \wedge P(z)]$.

So $Q(z) \wedge P(z)$ holds for all z in the domain D.

Now let c be some element in the domain D.

So $Q(c) \wedge P(c)$ holds (by instantiation), and therefore $Q(c)$ by itself holds.

But c could have been any element of the domain D.

So we conclude $\forall x.Q(x)$. (by generalization)

We conclude $\forall y.P(y)$ similarly (by generalization). Therefore,

$$\forall x.Q(x) \wedge \forall y.P(y) \qquad \text{QED.}$$

- Quantifiers
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Mathematical Proof

We prove mathematical statements by using logic.

$$\frac{P \to Q,\ \ Q \to R,\ \ R \to P}{P \wedge Q \wedge R}$$

not valid

To prove something is true, we need to assume some **axioms**!

This was invented by Euclid in 300 BC,
who began with 5 assumptions about geometry,
and derived many theorems as logical consequences.

http://en.wikipedia.org/wiki/Euclidean_geometry

Ideal Mathematical World

What do we expect from a logic system?

- What we prove is true. (soundness)
- What is true can be proven. (completeness)

Hilbert's program

- To resolve foundational crisis of mathematics (e.g. paradoxes)
- Find a finite, complete set of axioms,
  and provide a proof that these axioms were consistent.

http://en.wikipedia.org/wiki/Hilbert's_program

Power of Logic

Good news: Gödel's Completeness Theorem

Only need to know a few axioms & rules, to prove *all* validities.

That is, starting from a few propositional & simple predicate validities, every valid first order logic formula can be proved using just universal generalization and *modus ponens* repeatedly!

modus ponens

$$\frac{P \rightarrow Q, P}{Q}$$

Limits of Logic (Optional)

Gödel's *In*completeness Theorem for Arithmetic

For any "reasonable" theory that proves basic arithmetic truth, an arithmetic statement that is true, but not provable in the theory, can be constructed.

(very very brief) proof idea:

Any theory "expressive" enough can express the sentence

"This sentence is not provable."

If this is provable, then the theory is inconsistent.

So it is not provable.

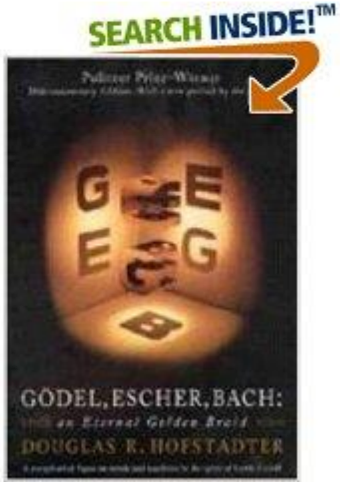
Limits of Logic (Optional)

Gödel's *Second Incompleteness* Theorem *for Arithmetic*

For any "reasonable" theory that proves basic arithemetic truth, it cannot prove its consistency.

No hope to find a complete and consistent set of axioms?!

An excellent project topic:

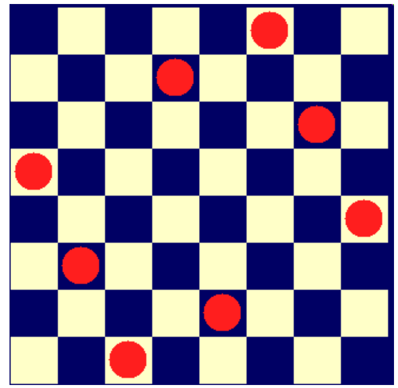


http://en.wikipedia.org/wiki/G%C3%B6del%27s_incompleteness_theorems

Applications of Logic

Logic programming

solve problems by logic

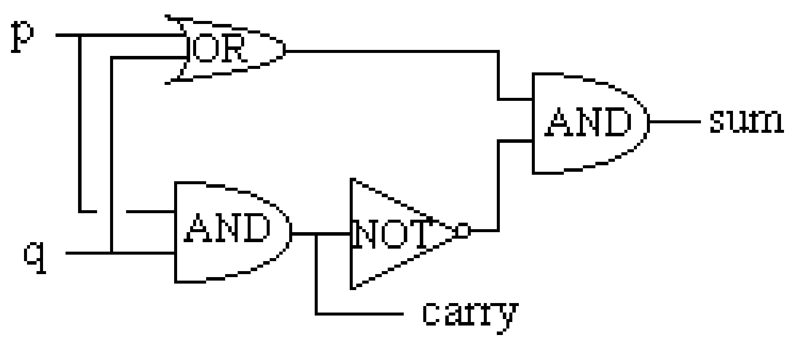


	1		6		7			4
	4	2						
8	7		3			6		
	8			7			2	
			8	9	3			
	3			6			1	
		8			6		4	5
						1	7	
4			9		8		6	

Database

making queries, data mining

Digital circuit



p	q	sum	carry
1	1	0	1
1	0	1	0
0	1	1	0
0	0	0	0

Summary

This finishes the introduction to logic, half of the first part.

In the other half we will use logic to do mathematical proofs.

At this point, you should be able to:

- Express (quantified) statements using logic formula
- Use simple logic rules (e.g. DeMorgan, contrapositive, etc)
- Fluent with arguments and logical equivalence